

HALLOWEEN V8

Title : IT LOG ANALYZER

Team Name : YOU DECIDE

Team ID : ACGC1B

Team Members : KHWAHISH PANDEY, SUBHAM

ROUT, HARSH KUMAR PANDEY, NIRUTH ANANTH

Problem Statement



IT SYSTEM LOG ANALYSER: The CRPF has deployed units and personnel in diverse locations across the country. However, there is currently no centralized system in place for experts to analyze IT system logs, evaluate potential threats, and identify breaches.

Log Analysis helps to discover many loopholes in networks that are opened intentionally by attackers or unintentionally because of misconfiguration, and bugs in the Software, Hardware, and Firmware.



Solution

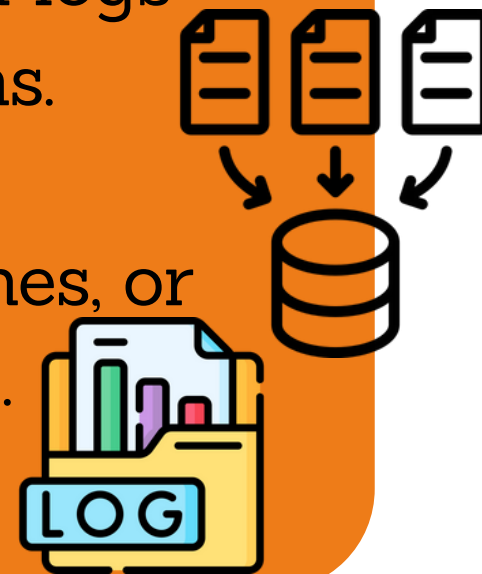


“LOG LENS” : We have developed a log analyzer integrating blockchain and machine learning for secure and tamper-proof log storage and efficient monitoring. Utilizing blockchain ensures the integrity of log records, making them immutable and resistant to manipulation or forgery

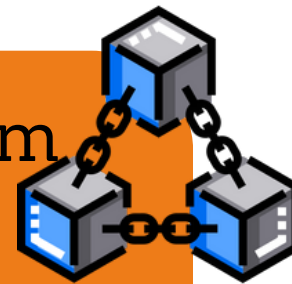


Collect and analyze system logs from distributed systems.

Detect cyber threats, breaches, or anomalies in real-time.



We employed InterPlanetary File System (IPFS) for decentralized storage in blockchain, eliminating failure points. Cryptographic hashes ensure data integrity, reducing redundancy and preserving decentralization for off-chain data.



The system offers interactive data visualization tools, including graphs and charts, to help experts gain insights from log data more effectively

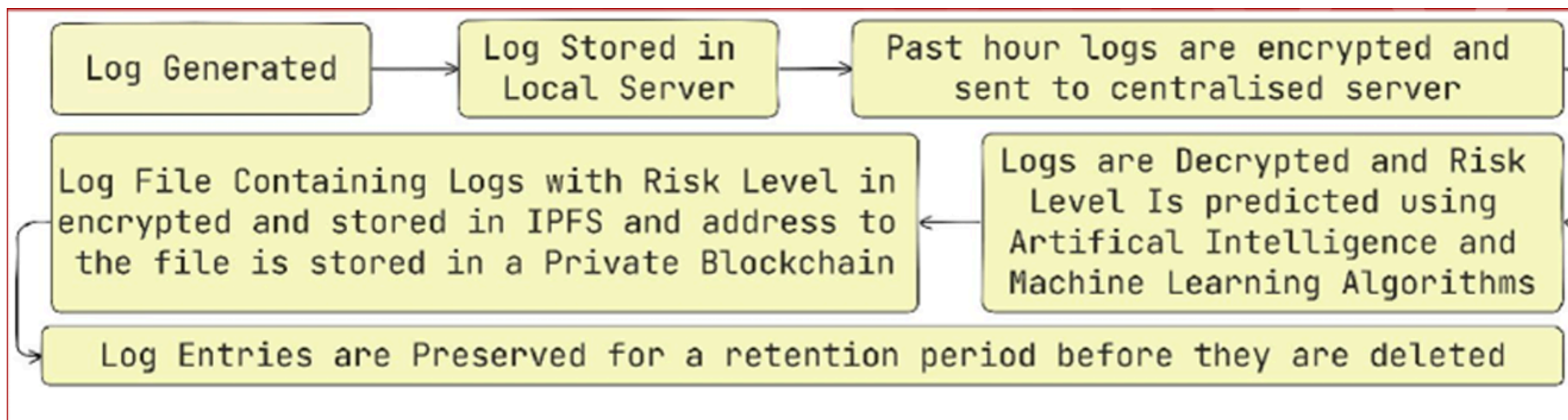


Tech & Implementation

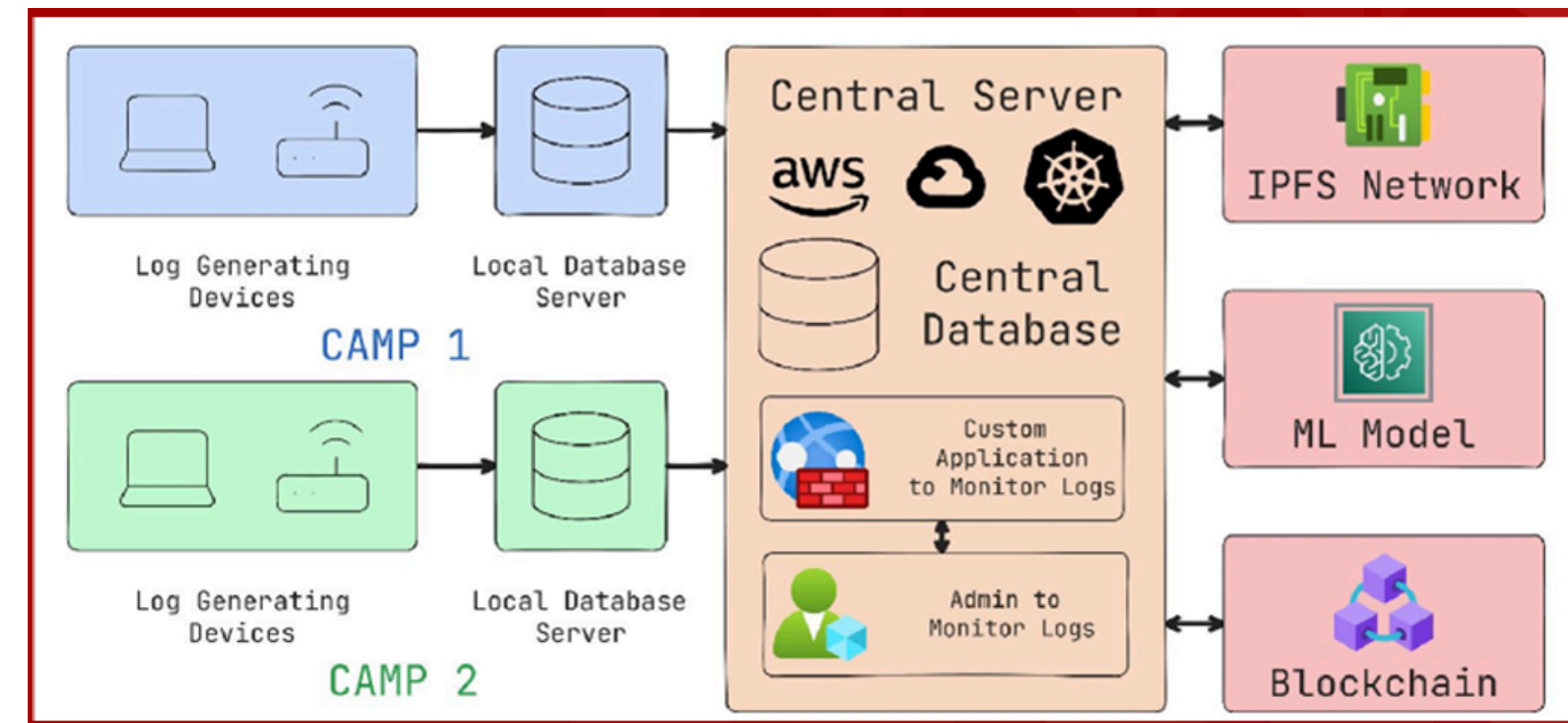
- Tech Stack Used :



- FlowChart :



- Infrastructure :

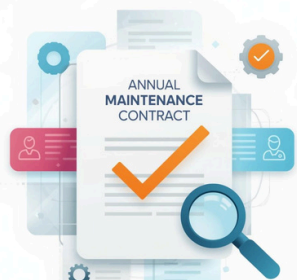


Business Model & Impact



Government Funded Deployment

This model involves financing from the CRPF or Ministry of Home Affairs, emphasizing national security and ensuring resource allocation for the cybersecurity dashboard platform's development and maintenance.



Software Licensing Model



Potential expansion to other agencies, such as BSF and CISF, allows for revenue generation through software licenses, enhancing collaborative cybersecurity efforts across various sectors of national security.



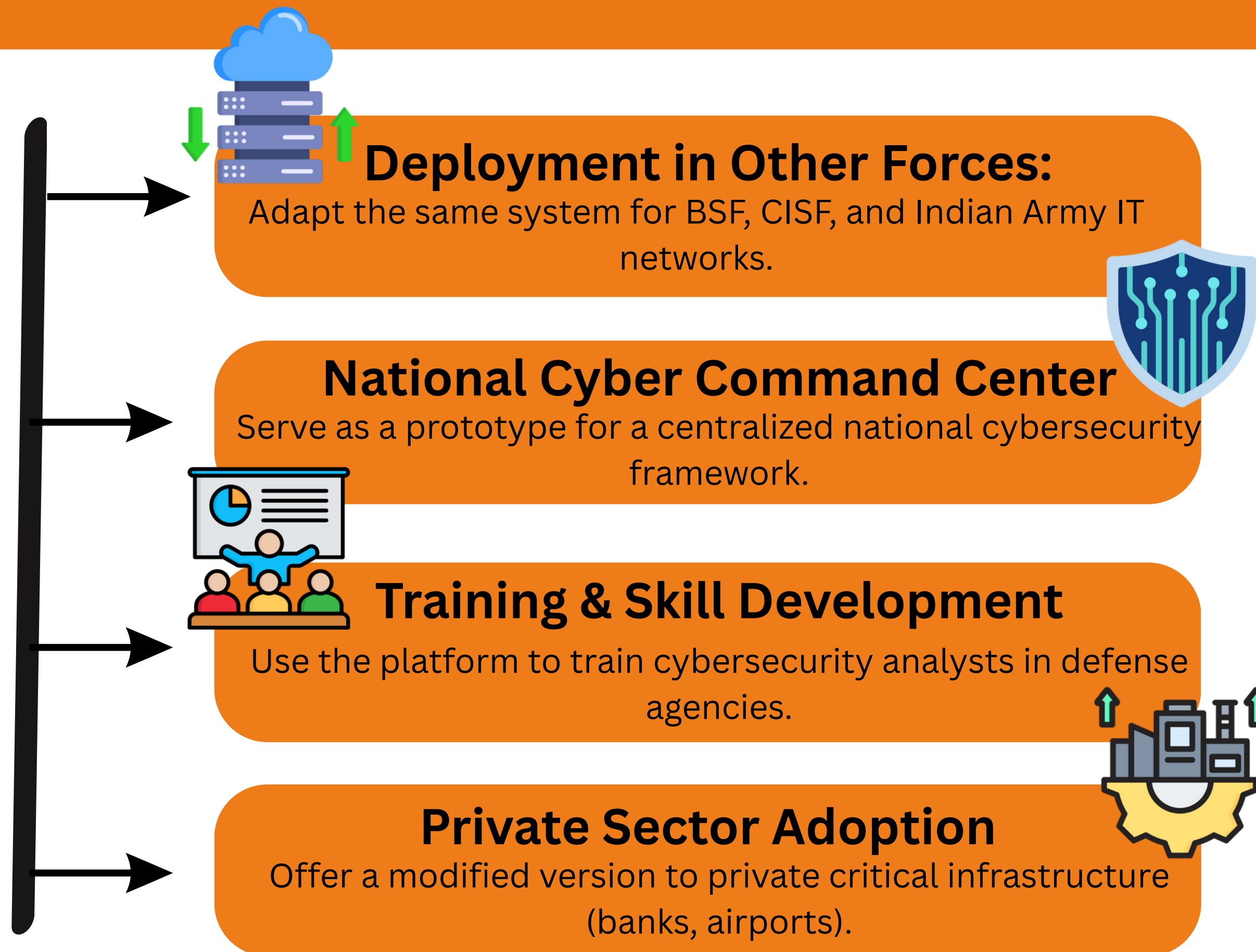
Impact

Reduces the chances of data breaches, defacement, and espionage, strengthening digital defense across CRPF infrastructure

🧠 Impact: Proactive threat detection prevents damage before it occurs — turning cybersecurity from reactive to preventive.



Future Scope & Conclusion



Thank
You

